Clinical Study Summary Report (CSSR) Document Status: Draft | **Version:** 1.0 | **Date:** 03-Apr-2026

---

1. Administrative & Study Identification

Study Title: A Study of Sotatercept for the Treatment of Cpc-PH Due to HFpEF (MK-7962-007/A011-16) **Official Title:** A Phase 2, Double-blind, Randomized, Placebo-controlled Study to Evaluate the Effects of Sotatercept Versus Placebo for the Treatment of Combined Postcapillary and Precapillary Pulmonary Hypertension (Cpc-PH) Due to Heart Failure With Preserved Ejection Fraction (HFpEF) **Short Title/Acronym:** CADENCE **Protocol Number:** MK-7962-007 / A011-16 **NCT Number:** NCT04945460 **Sponsor / Company:** Merck **Trial Status:** ACTIVE_NOT_RECRUITING (Active but not currently recruiting) **Investigational Product (Drug Name):** Sotatercept (Drug Preferred Name: Sotatercept) **Phase of Development:** PHASE2 **Targeted Enrollment:** 164 participants **Medical Monitor:** [Name and Contact Information]

---

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the efficacy, safety, and tolerability of sotatercept versus placebo in adults with Cpc-PH due to HFpEF. Efficacy is measured by the change from baseline in pulmonary vascular resistance (PVR) at Week 24. **Secondary Objectives:** To assess changes in 6-minute walk distance (6MWD, key secondary endpoint), time to first clinical worsening event (TTCW), and mean pulmonary arterial pressure (mPAP).

2.2 Background & Rationale To address the unmet medical need in patients with combined pre- and postcapillary pulmonary hypertension (Cpc-PH) and heart failure with preserved ejection fraction (HFpEF). Currently, there are no approved treatments specifically for CpcPH-HFpEF. Sotatercept, a novel activin signaling inhibitor, acts as a ligand trap to restore the balance between pro-proliferative and anti-proliferative signaling pathways. This mechanism has the potential to reverse pathological vascular changes and improve hemodynamics in this distinct patient population.

---

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Placebo-controlled **Blinding:** Double-blind **Randomization:** 1:1:1 (Placebo, 0.3 mg/kg sotatercept, 0.7 mg/kg sotatercept)

3.2 Planned Enrollment & Duration Targeted Enrollment: 164 participants **Number of Sites:** Global and Regional multicenter locations **Total Duration:** Up to 114 weeks (28-day Screening

Period, 24-week double-blind placebo-controlled Treatment Period, and an 8-week Follow-up Period. *Note: As of protocol amendment 6, the 18-month Extension Period is being removed*).

---

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age 18 to 85 years. 2. Clinical diagnosis of HFpEF (Left ventricular ejection fraction ≥50%, with no history of LVEF below 45% in more than two consecutive measurements under stable conditions). 3. Demonstrated Cpc-PH by baseline RHC with a minimum PVR of ≥320 dyn•sec/cm5 (4 wood units), mean pulmonary arterial pressure (mPAP) of >20 mmHg, and pulmonary capillary wedge pressure (PCWP) >15 mmHg but <30 mmHg. 4. New York Heart Association (NYHA) functional class (FC) II or III. 5. Six-minute Walk Distance (6MWD) ≥100 m repeated twice during Screening. 6. Adherence to highly effective contraception rules for women of childbearing potential and male participants during the study and for 16 weeks (112 days) after discontinuation.

4.2 Exclusion Criteria 1. Diagnosis of PH in WHO Group 1, WHO Group 3, WHO Group 4, or WHO Group 5. 2. Clinically significant and active lung disease (e.g., COPD with FEV1 <60% predicted, restrictive lung disease with TLC <70% predicted). 3. Occurrence of myocardial infarction, acute coronary syndrome, CABG, or PCI within 180 days, or stroke within 90 days of Visit 1. 4. Acutely decompensated HF requiring hospitalization within 30 days of Visit 1. 5. Received approved PAH-specific therapies (e.g., ERAs, prostacyclin analogs, PDE-5 inhibitors) within 30 days of Visit 1. 6. Received intravenous inotropes within 30 days or erythropoietin within 6 months of Visit 1. 7. Uncontrolled systemic hypertension (>160 mmHg systolic or >110 mmHg diastolic) or systemic hypotension (<90 mmHg systolic or <50 mmHg diastolic). 8. Resting heart rate of <45 bpm or >115 bpm. 9. Body mass index ≥50 kg/m2.

---

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Placebo, 0.3 mg/kg sotatercept, or 0.7 mg/kg sotatercept administered during the placebo-controlled Treatment Period. **Route of Administration:** Subcutaneous injection. **Duration of Treatment:** 24-week double-blind, placebo-controlled treatment period.

5.2 Concomitant & Prohibited Therapy Permitted: Chronic medication for HF or underlying conditions at a stable dose for ≥30 days prior to Visit 1. Diuretics/anticoagulants are excepted but should not be newly started/stopped within 30 days. Oral PDE-5 inhibitors for erectile dysfunction are permitted if not given within 48 hours of a study visit. **Prohibited:** Concurrent investigational therapies (small molecules within 30 days; biologics within 5 half-lives).

---

6. Study Timeline & Visit Schedule

| Visit ID | Window (Days) | Key Activities/Procedures | | :--- | :--- | :--- | | **Screening** | Day -28 to 0 | Informed Consent, Medical History, Baseline Hemodynamics (RHC), 6MWD | | **Baseline** | Day 0 | Randomization, First Dose Administration, Vitals | | **Interim** | Q3W | Dosing, Safety Labs, Efficacy Assessment, Adverse Event Monitoring | | **End of Study** | Week 24 | Final Hemodynamics via RHC, 6MWD, IP Accountability |

(Note: The trial incorporates an 8-week safety Follow-up Period post-treatment)

---

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint * **Change From Baseline in Pulmonary Vascular Resistance (PVR) at Week 24:** PVR, a hemodynamic variable of pulmonary circulation, is measured by right heart catheterization (RHC).

7.2 Secondary Endpoints * **Change From Baseline in the 6-Minute Walk Distance (6MWD) at Week 24** * **Time to First Clinical Worsening Event (TTCW) at Week 24** * **Change From Baseline in Mean Pulmonary Arterial Pressure (mPAP) at Week 24**

---

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard clinical collection. Frequent AEs evaluated include bleeding, diarrhea, fatigue, increased hemoglobin, peripheral edema, and dizziness. **Serious Adverse Events (SAEs):** Reported promptly per standard regulatory protocol guidelines. **Data Monitoring Committee (DMC):** Independent safety monitoring mechanisms managed by the sponsor.

---

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) for primary efficacy; Safety Set for all AE/SAE evaluations. **Sample Size Justification:** Target enrollment of 164 participants designed to power primary PVR hemodynamic endpoint shifts against placebo at $\alpha = 0.05$. **Handling of Missing Data:** Standard multiple imputation (MI) or Mixed-Effect Model Repeat Measurement (MMRM) for continuous measures (PVR, 6MWD).

---

10. Quality Assurance & Regulatory Compliance

GCP Compliance: Conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Phase Requirement Criteria:** PHASE_REQUIREMENT (Criteria text: Trial must be in PHASE2. Phase ID: PHASE2). **Monitoring Plan:** Routine onsite and remote risk-based monitoring per sponsor protocol. **Audit Trail:** eCRF locking and data clarification forms via secure clinical data management systems.

---

11. Study Identifiers & Governance

Primary ID: MK-7962-007 / A011-16 **Posting ID:** 339426069472573451 **Registry Number:** NCT04945460 **Sponsor / Company:** Merck **Status Name:** ACTIVE_NOT_RECRUITING **Status Description:** Active but not currently recruiting **Phase ID:** PHASE2

---

12. Clinical Framework & Disease Classifications

Disease Areas: Hypertension, Pulmonary **Name:** Hypertension **Preferred Name:** Hypertensive disease **Preferred UMLS Name:** Combined pre- and postcapillary pulmonary hypertension **Semantic Type:** Disease or Syndrome **Definition:** The presence of chronic increased pressure in the systemic arterial system. **Ontology / Mapping Codes:** * **MeSH Code:** D006973 * **HPO Codes:** HP:0000822 * **SNOMED ID:** 38341003 (Hierarchy: 49601007|Disorder of cardiovascular system; 64572001|Disease) **Optimization Target:** Reduction in Pulmonary Vascular Resistance (PVR) and improvement in functional capability (6MWD).

---

13. Study Procedures & Methodology

Management Pathway: Standard clinical assessment involving regular Right Heart Catheterizations (RHC) and functional tests. **Education Protocol (HCP):** Protocol training, safety/bleeding management, and injection procedures. **Education Protocol (Patient):** Trial compliance, adverse event reporting instructions, and Q3W onsite administration tracking. **Quality Audit Mechanism:** Independent central reading of RHC hemodynamic outputs and site monitoring.

---

14. Observation & Follow-Up Schedule

Total Duration: Up to 114 weeks (Screening, 24-week double-blind treatment, and 8-week follow-up segments). **Onsite Visit Cadence:** Every 3 weeks (Q3W) aligning with the dosing schedule. **Remote Monitoring Frequency:** As required between clinical visits.

Checklist Review Interval: End of the 24-week double-blind period for primary endpoint analysis.

---

15. Sign-off & Approval

Role	Name	Signature	Date
Principal Investigator	[Name]	____	[DD-MMM-YYYY]
Clinical Operations Lead	[Name]	___	[DD-MMM-YYYY]
Lead Statistician	[Name]	_____	[DD-MMM-YYYY]